Conduction, convection, radiation

antocciphysics | TEACHER KEY | CPA

Learning target: Identify mechanisms of heat transfer and support each classification.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 22; 3 course days later.*
5 W then 2.5 W. Same energy over twice the time gives half the power.

02. Explain a worked example (worked)
Conduction via microscopic interactions; convection via bulk fluid motion; radiation via electromagnetic waves.

03. Look and explain (model)
Conduction; convection; radiation. Accept noting multiple effects if primary mechanism is justified.

04. Build the solution (practice)
Bulk fluid movement versus microscopic transfer in a solid; buoyancy-driven loop in stated heating arrangement.

05. Use the idea (reason)
Radiation; no surrounding air for convection in the near-vacuum model.

06. Choose a method (mixed) *Review: Unit 4, Day 13; 12 course days later.*
Difference 8 J may be rotational energy and thermal energy; other justified transfers acceptable.

07. Exit check (exit)
Electromagnetic radiation crosses vacuum.

08. STRETCH: Apply it another way (stretch)
Examples low-conductivity wall, lid limiting fluid circulation/evaporation, reflective surface reducing radiative transfer; explain limits.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 22; 3 course days later.*
v_{escape}=10 m/s. Assumptions include ideal central gravity, no atmosphere, no further thrust and negligible other bodies.

02. Explain a worked example (worked)
Conduction via microscopic interactions; convection via bulk fluid motion; radiation via electromagnetic waves.

03. Represent and interpret (model)
Conduction; convection; radiation. Accept noting multiple effects if primary mechanism is justified.

04. Solve and justify (practice)
12, 000 J; a larger thickness lengthens the conduction path; ideal steady conduction rate inversely proportional to thickness.

05. Analyze the reasoning (reason)
Radiation; no surrounding air for convection in the near-vacuum model.

06. Choose a method (mixed) *Review: Unit 4, Day 13; 12 course days later.*
mgh≈2.352 J; K=0.8 J; fraction≈0.3401 (34.01%).

07. Exit check (exit)
Net thermal radiation; any cooler radiative environment can receive net energy, not only 0 K.

08. STRETCH: Apply it another way (stretch)
Conduction through supports/neck and radiation remain; compare neck insulation and surface emissivity while controlling temperatures. Vacuum suppresses fluid conduction/convection, not all transfer.